

# Learning the Effective PSF: An Amortized, Neural PSF Field for DES Single-Epoch Imaging

JEFFREY REGIER<sup>1</sup>

(THE LSST DARK ENERGY SCIENCE COLLABORATION)

<sup>1</sup>*Department of Statistics, University of Michigan, Ann Arbor, MI 48109, USA*

## ABSTRACT

Accurate point-spread-function (PSF) models are the dominant systematic in weak-lensing shear measurement. Per-exposure fitters (e.g., PSFEX, PIFF) represent the PSF as a sub-pixel surface-brightness profile that must be convolved with an assumed pixel response to predict an image. We instead target the *effective* PSF—the pixel-convolved point-source response, i.e. the expected pixel values a point source produces. This is the quantity that sampled data actually identify and the exact kernel needed to forward-model galaxies. A single *amortized* neural network (multi-head attention over an exposure’s stars—position, color, flux, and pixel cutouts) predicts this effective PSF at any queried position and oversampling, with no per-exposure fitting at test time; trained by inverse-variance least squares, it regresses the conditional expectation of the pixels, so its training target, the quantity it models, and the reserved-star residuals it is validated on all coincide. On 3,301 frozen test exposures of Dark Energy Survey (DES) single-epoch imaging (CCD 31, *grizY*; 73,609 reserved stars), the model matches or exceeds PIFF and PSFEX on every metric: size-residual scatter 0.030 versus 0.039/0.037, ellipticity errors  $|\delta e_1|, |\delta e_2| = 0.015/0.015$  versus 0.017/0.016 (PIFF), and shape correlations 0.65/0.61 versus 0.61/0.58 (PIFF), at  $\chi^2/\text{dof} = 1.04$ . In a star-anchored galaxy injection–recovery test on real data, our PSF recovers galaxy *ellipticity* without coherent bias ( $|\Delta e_1| \lesssim 0.01$ , on par with PIFF)—the weak-lensing-critical quantity—while recovered galaxy *sizes* run  $\sim 8\%$  small. We trace this to the coordinate decoder under-resolving the sharp PSF core and reduce it with a sharper frequency encoding. Color conditioning removes a chromatic PSF-size systematic that the baselines retain (+0.004/mag versus  $\sim -0.06/\text{mag}$  residual slope). The model is flat in the number of available fit stars where PIFF degrades sharply ( $\chi^2 \simeq 1.1$  versus 5.1 at five stars). Using simulations as a controlled instrument, we show that a *blend forward-model* training loss removes a size bias from unrecognized blends, and we identify and fix a spin-2 representation failure in which the network corrupts the PSF ellipticity by attending to galaxy detections in its context; restricting attention to point sources recovers it fully.

**Keywords:** Weak gravitational lensing (1797) — Point spread function (1349) — Astronomical techniques (1684) — Convolutional neural networks (1938) — Sky surveys (1464)

## 1. INTRODUCTION

Weak gravitational lensing infers the matter distribution from coherent  $\sim 1\%$  distortions of galaxy shapes. The measurement is limited by knowledge of the PSF: any error in the modeled PSF size or ellipticity propagates directly into the inferred shear. Survey pipelines model the PSF per exposure by fitting that exposure’s stars with a flexible model interpolated across the focal plane (PSFEX; E. Bertin 2011) (PIFF; M. Jarvis et al. 2021). This is robust but (i) discards information shared across exposures and nights, (ii) degrades when few clean stars are available, and (iii) is purely empir-

ical at the star positions—it does not exploit auxiliary information such as stellar color, which sets the chromatic PSF.

We take an amortized, learning-based approach. Rather than fit each exposure’s PSF from scratch, a single network learns—across many exposures—a *data-adaptive prior* over how the PSF looks and varies, and applies it to a new exposure in one forward pass, mapping that exposure’s stars (the *context*) to a continuous PSF field queryable at arbitrary position, color, flux, and resolution. Amortizing a learned prior in place of per-exposure curve fitting is the central idea: it lets the

model pool information across the training set that no single-exposure fit can access (and, in principle, across the focal plane). The inferential target is the PSF at *star-free* positions, where galaxies are measured; predicting held-out stars is the training objective and the validation instrument. We train and evaluate on DES single-epoch data and benchmark against PIFF and PSFEX run by us under matched conditions.

A second, more fundamental difference is the *target* of the model. Traditional PSF models—including PSFEX (super-sampled pixels) and PIFF (a surface-brightness profile on the sky)—represent the PSF as a continuous function on  $\mathbb{R}^2$  at finer-than-pixel resolution, which must then be convolved with an assumed pixel-response kernel to predict what a detector records. We instead model the *effective* PSF (ePSF; J. Anderson & I. R. King 2000): the instrumental PSF already integrated over the pixel, i.e. the expected fraction of a point source’s light falling in each pixel as a function of the source’s sub-pixel position. The network maps a query position (and color, flux) directly to these expected pixel values, never representing the continuous PSF explicitly—the sense in which our PSF is *implicit*. This target is more principled on two grounds. First, identifiability: from sampled data one cannot uniquely recover the sub-pixel surface brightness without extra assumptions, whereas the ePSF is exactly what the pixels constrain—the reason it underpins state-of-the-art point-source astrometry and photometry (J. Anderson & I. R. King 2000; T. R. Lauer 1999; T. I. Liaudat et al. 2023). Second, sufficiency: the ePSF as a function of source position is a sufficient statistic for forward-modeling *any* source—a galaxy’s expected pixel data is its surface brightness convolved with the ePSF, with no separate pixel step (G. M. Bernstein & M. Jarvis 2002)—so the ePSF is precisely the kernel a shear pipeline needs. The only residual is discretizing that convolution on a finite grid, which our ability to render at arbitrary oversampling controls directly.

## 2. METHOD

*Architecture*—Each detected source is encoded from its pixel cutout (a learned image encoder applied to the scale-normalized stamp), its sky position (sinusoidal encoding), and scalar color and log-flux. A multi-head self-attention block (A. Vaswani et al. 2017) lets every query attend to the context sources; a coordinate decoder then renders the PSF at the query position and any oversampling factor, returning the pixel-convolved effective PSF used to convolve galaxy models. We use a polar (spin-2-aware) angular parametrization of the

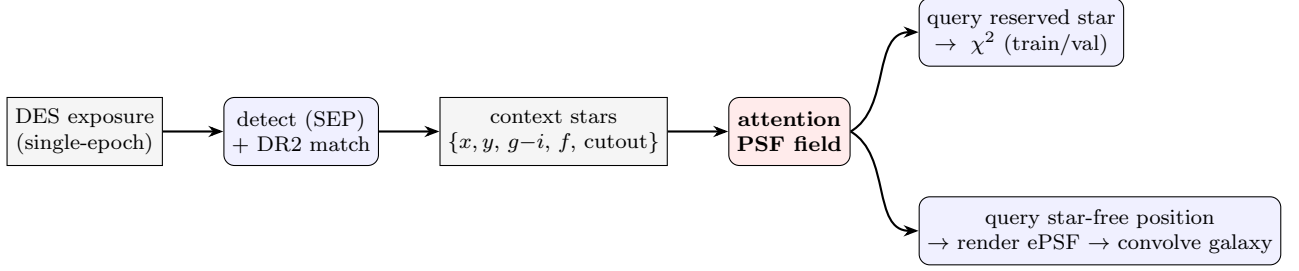
decoder and FiLM modulation of the decoder by the attended context features.

*Inferential target (effective PSF)*—The decoder outputs the effective PSF (J. Anderson & I. R. King 2000)—the pixel-integrated point-source response—sampled at the query grid, optionally oversampled. Because the loss below compares predicted stamps to *observed* star pixels, the model regresses the observable by construction: it learns  $\mathbb{E}[\text{pixel value} \mid \text{point source at } (x, y)]$  directly, rather than fitting a sub-pixel profile and assuming a pixel kernel. Identifiability aligns the three things that a non-identifiable model keeps separate: our training target, the quantity we model, and the quantity we validate all coincide, so reserved-star residuals (§4) directly test what we fit. A surface-brightness model is instead validated only through pixel- or moment-space summaries of its rendered output—the same observable space—so its extra sub-pixel degrees of freedom are never themselves constrained by the data. This is also what makes the benchmark fair: we score all three methods on moments measured in pixel space (model stamp versus star stamp), so no method is privileged by its internal representation. The same pixel-convolved output is the kernel used to forward-model galaxies in §4.4; oversampling the render reduces the discretization error of that convolution for compact sources.

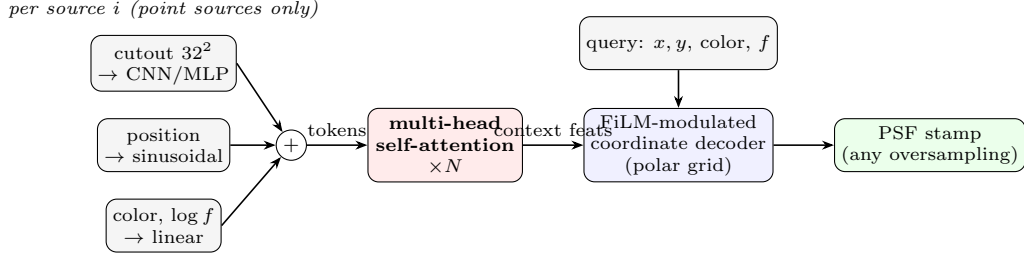
*Loss*—Training minimizes an inverse-variance  $\chi^2$  between predicted and observed star cutouts, with the per-star amplitude solved in closed form. Star pixels are noisy realizations whose conditional mean is the effective PSF; a squared-error objective is minimized by the conditional expectation  $\mathbb{E}[\text{pixel} \mid \text{source, context}]$ , so with enough capacity and data the objective is minimized by that quantity—the identifiable effective PSF. The network thus targets only what the pixels constrain, rather than a finer-than-pixel profile the sampled data do not determine. We introduce a *blend forward-model* loss: rather than treat each clean star as isolated, we model its cutout as a generalized-least-squares sum of the target plus its detected neighbors, all rendered by the same decoder (§4.7). Detected galaxies near a target are handled by exclusion (default), masking, or a crude extended-component model; the three are within noise on real reserved stars.

*Point-source context*—The PSF should be informed only by point sources. We restrict the attention context to stellar detections; §4.7 shows this is necessary to avoid a spin-2 corruption from extended sources in the context.

*Color conditioning*—The decoder conditions on  $g-i$  color, enabling a per-object chromatic PSF (differen-



**Figure 1.** Overall workflow. A single network, trained across exposures, maps an exposure’s detected stars (context) to a continuous PSF field. Querying held-out reserved stars provides the training objective and validation; the inferential target is the PSF at star-free positions, rendered (oversampled) to convolve galaxy models.



**Figure 2.** Network architecture. Each context source (restricted to point sources; §2) is encoded from its pixel cutout, sky position, and scalar color/log-flux into a token. Multi-head self-attention mixes the context; a FiLM-modulated coordinate decoder, conditioned on the attended features and the query (position, color, flux), renders the pixel-convolved PSF at any position and oversampling factor.

tial chromatic refraction and wavelength-dependent seeing) that per-exposure empirical models cannot represent (§4.5).

### 3. DATA

We use DES single-epoch FITS (Y5A1, CCD 31; amplifier B is dead, masked via MSK bit 8). Sources are detected with SEP and cross-matched to DES DR2 for colors, SPREAD\_MODEL star/galaxy labels, and coadd object IDs. Splits are at the observing-night level by deterministic hash to prevent leakage (consecutive exposures share atmosphere), frozen in a committed manifest with  $\sim 20\%$  of clean stars per test/validation exposure reserved for scoring only; the test split is 3,301 exposures across *grizY*. Reserved stars are excluded from every method’s fit/context and scored identically across methods on the same pixel grid.

*Simulations*—A matched pipeline renders Moffat PSF fields (second-order polynomial FWHM and shear variation—exactly representable by PIFF/PSFEX, a deliberately fair-to-baselines choice) into the same data format, optionally with chromatic FWHM—color dependence, blended stars, and injected galaxy detections. Star density, galaxy density (3:1), and per-source S/N are tuned to match the real extraction. The simulation is the controlled instrument for the systematics studies in §4.7.

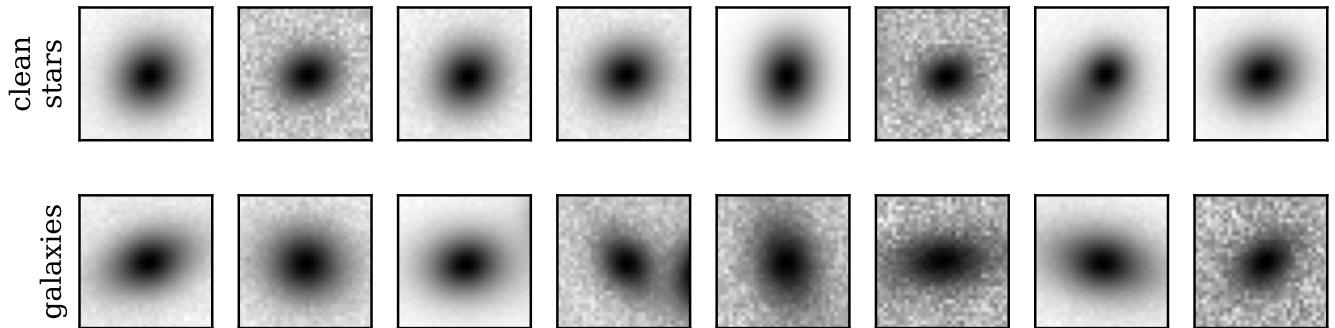
## 4. RESULTS

### 4.1. Reserved-star accuracy on real data

On 3,301 test exposures and 73,609 reserved stars, the amortized model (no per-exposure fitting) matches or exceeds both baselines on every metric (Table 1). Reserved “clean” stars carry a mild blend-contamination floor that suppresses absolute correlations for *all* methods; the comparison is fair (identical stars) and the ordering is robust.

### 4.2. Pixel-level and astrometric validation

Because the model regresses the effective PSF—the expected pixel values—the reserved-star quality of fit can be assessed directly in the modeled (pixel) space, the validation native to effective-PSF work (J. Anderson & I. R. King 2000). Two checks follow. First, the inverse-variance  $\chi^2/\text{dof}$  is a noise-calibrated goodness of fit: at 1.04 (Table 1) the per-pixel residuals are consistent with photon and read noise, i.e. no unmodeled pixel-level structure—a statement available to a sub-pixel model only through its rendered output. The variance is the survey’s own inverse-weight (photon plus read noise) map, not a quantity tuned to the residuals, so  $\chi^2 \approx 1$  is an independent calibration check rather than a fitted outcome. Second, the predicted ePSF centroid matches the star centroid to 0.010 px (2.6 mas) in median offset, versus 0.017 px (4.5 mas) for PIFF and

Example simulated cutouts ( $32 \times 32$  px, arcsinh stretch)

**Figure 3.** Example simulated cutouts. Clean stars (top) are compact point sources; injected galaxies (bottom) are visibly extended and elliptical (second-moment size  $T \approx 33$  vs  $15 \text{ px}^2$ ). Both are presented to the network as context.

**Table 1.** Reserved-star residuals on the DES test split (3,301 exposures, 73,609 stars).  $T$  scatter is the robust standard deviation of  $\delta T/T$ ;  $|\delta e|$  are median absolute ellipticity residuals; corr is the model–star shape correlation. Best in bold.

| Method           | $T$ scat      | $ \delta e_1 $ | $ \delta e_2 $ | corr( $e_1$ ) | corr( $e_2$ ) | $\chi^2/\text{dof}$ |
|------------------|---------------|----------------|----------------|---------------|---------------|---------------------|
| <b>This work</b> | <b>0.0301</b> | <b>0.0151</b>  | <b>0.0150</b>  | <b>0.650</b>  | <b>0.607</b>  | 1.037               |
| PIFF             | 0.0393        | 0.0171         | 0.0158         | 0.611         | 0.577         | 1.073               |
| PSFEx            | 0.0371        | 0.0160         | 0.0154         | 0.645         | 0.604         | 1.047               |

**Table 2.** Astrometric residual of the model PSF centroid relative to the star, on the reserved test stars. Per-axis scatter is the robust standard deviation; the offset is the median radius ( $0.263''/\text{px}$ ). Best in bold.

| Method           | $\sigma(\delta x)$ [px] | $\sigma(\delta y)$ [px] | median $ \delta $ [mas] |
|------------------|-------------------------|-------------------------|-------------------------|
| <b>This work</b> | <b>0.008</b>            | <b>0.009</b>            | <b>2.6</b>              |
| PIFF             | 0.015                   | 0.013                   | 4.5                     |
| PSFEx            | 0.015                   | 0.017                   | 5.1                     |

0.020 px (5.1 mas) for PSFEx (Table 2); all three centroids are measured with the same adaptive-moment routine on the same pixel grid, so the difference reflects the models, not a centroiding convention. This is a near  $2\times$  improvement on a quantity the ePSF formalism is built to optimize.

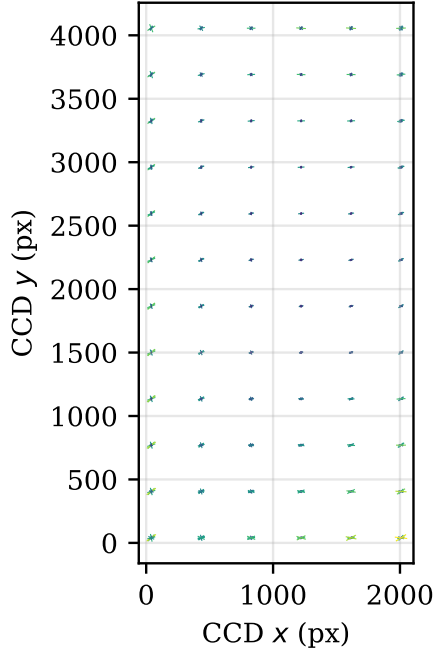
Third, stacking the unit-normalized residual (observed star minus model, as a fraction of total light) over 306 reserved stars isolates any coherent shape error that second moments can miss (Figure 6). Our model leaves the flattest residual (RMS  $2.6 \times 10^{-4}$ ); PIFF and PS-

FEx leave a visible negative core ( $3.5$  and  $4.4 \times 10^{-4}$ ), a higher-order mismatch in the central pixels that is consistent with their larger size residuals and is invisible to the size and ellipticity summaries alone.

#### 4.3. Residual two-point statistics

Per-star residuals understate the impact on weak lensing, where the *spatially coherent* part of the PSF-modeling error sources additive shear bias. We therefore compute the standard  $\rho$ -statistics (B. Rowe 2010; M. Jarvis et al. 2016) of the reserved-star residuals—the auto- and cross-correlations of the residual ellipticity  $\delta e$  and the size-leakage term  $e \delta_T$ —accumulated within exposures with TREECORR (Figure 7). On the same reserved stars used for Table 1, the amortized model is at or below both per-exposure baselines on four of the five statistics: the median  $|\xi_+|$  across angular scales is  $3.9 \times 10^{-5}$  ( $\rho_1$ ),  $5.2 \times 10^{-5}$  ( $\rho_2$ ),  $2.5 \times 10^{-6}$  ( $\rho_3$ ), and  $4.8 \times 10^{-6}$  ( $\rho_4$ ), against  $\rho_1, \rho_2 \gtrsim 7 \times 10^{-5}$  and  $\rho_3 \simeq 6 \times 10^{-5}$  for PIFF. The strong  $\rho_3$  result—more than an order of magnitude below PIFF—reflects the model’s smaller and less spatially structured size residuals. The one exception is  $\rho_5 = \langle e(e \delta_T) \rangle$ , where our model ( $5.4 \times 10^{-5}$ ) sits above the baselines ( $\sim 10^{-5}$ ): a

True PSF ellipticity field (simulation)



**Figure 4.** The true, spatially varying PSF ellipticity field in a simulated exposure (whisker length  $\propto |e|$ ); the network must recover this at star-free positions.

residual coupling between PSF shape and the size error that a weak-lensing analysis would carry in its additive-bias budget. Overall the coherent, lensing-relevant part of the residual is no larger than that of the per-exposure baselines, despite the amortized model fitting no per-exposure free parameters.

#### 4.4. Galaxy recovery on real data

We test the inferential goal directly. We inject GALSIM galaxies, convolved with a strictly isolated reserved star’s empirical PSF, at that star’s position; each method predicts the PSF there from the *other* stars and recovers the galaxy. The truth arm (the anchor’s own image) recovers near-zero ellipticity bias, validating the protocol. On 32  $r$ -band test exposures (170 anchors; Table 3), all three methods recover galaxy *ellipticity* with low bias ( $|\Delta e_1| \lesssim 0.01$ ): the pixel-convolved effective PSF carries the PSF ellipticity faithfully into the forward model, so no method imprints a coherent shape error on the recovered galaxies—the weak-lensing-critical property, which our model satisfies on par with the per-exposure baselines. The methods separate instead on recovered *size*: PIFF is slightly large (+1.1%), while our recovered sizes are  $\sim 8\%$  small. This is *not* a second-moment size error—rendered at a fiducial flux, the model’s PSF size is accurate at both star and star-

**Table 3.** Star-anchored galaxy injection–recovery (32  $r$ -band test exposures, 170 anchors). The truth arm validates the protocol. All methods recover ellipticity with low bias; they separate on recovered size.

| Arm                | Size bias | $\Delta e_1$ | $\Delta e_2$ |
|--------------------|-----------|--------------|--------------|
| Truth (validation) | −1.2%     | −0.000       | −0.011       |
| <b>This work</b>   | −7.6%     | −0.001       | −0.008       |
| PIFF               | +1.1%     | −0.004       | −0.012       |

free positions (within 1%)—but a profile-shape effect: the model’s PSF is slightly *under-concentrated*, carrying  $\sim 2\%$  less encircled energy in the central few pixels than the empirical PSF, which biases recovered galaxy sizes low. We attribute this to spectral bias of the coordinate decoder, which under-resolves the sharp PSF core; §5 diagnoses this and shows it shrinks with a sharper coordinate encoding. The size deficit is the main residual systematic of our model; recovered ellipticity, the consequential quantity for shear, is unbiased.

#### 4.5. Chromatic PSF correction

In a chromatic simulation (FWHM varies  $-3\%/mag$  of  $g-i$ ), color conditioning reduces the residual chromatic size slope from  $-0.06/mag$  (PIFF, PSFEx, and our own zero-color ablation) to  $+0.004/mag$ —a  $\sim 17\times$  reduction. On real  $r$ -band data the paired color-versus-zero-color slope difference is  $+0.0024/mag$  (CI  $[+0.0016, +0.0030]$ ); the baselines structurally retain the systematic. This is information unavailable to a per-exposure empirical PSF.

#### 4.6. Sample efficiency

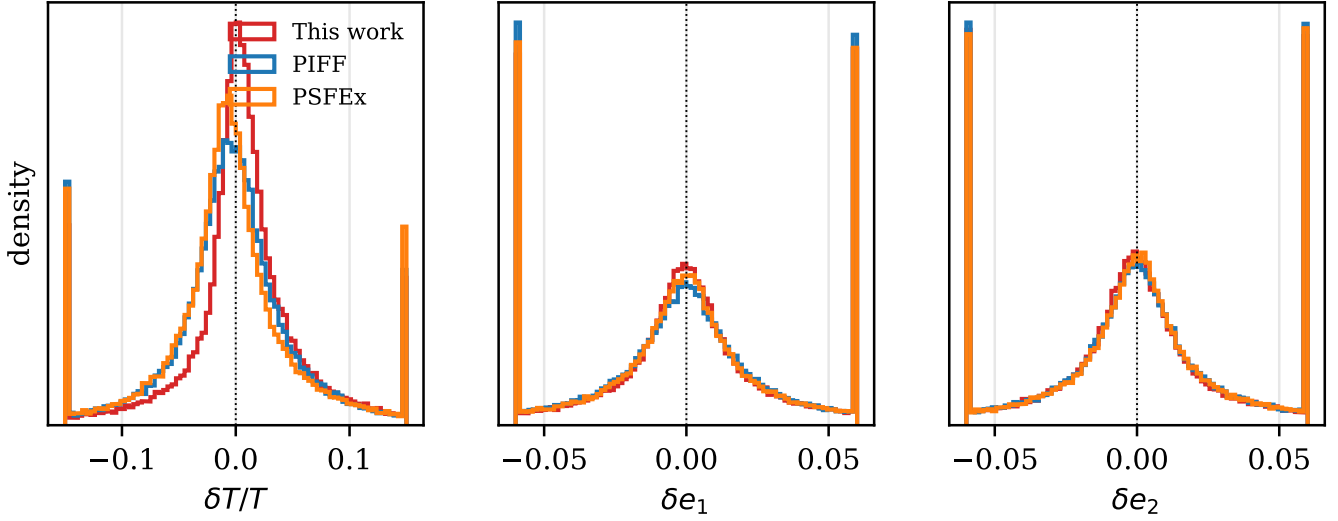
Restricting every method to  $k$  randomly chosen fit stars, the amortized model is approximately flat in  $k$  (size scatter  $0.078 \rightarrow 0.056$ ,  $\chi^2 \sim 1.1$  from  $k = 5$  to  $k = 100$ ), while PIFF collapses below  $k \approx 25$  ( $\chi^2 = 5.1$ , scatter 0.18 at  $k = 5$ ). Amortization transfers information from the training set to sparsely-populated exposures.

#### 4.7. Simulation studies

*Architecture ladder*—On the star-free truth grid,  $\chi^2/dof$  improves along additive  $\rightarrow$  FiLM  $\rightarrow$  diagonal  $\rightarrow$  polar ( $8.9 \rightarrow 4.96 \rightarrow 4.41$ ; PIFF 2.67; verified floor 1.0), recovering both ellipticity components.

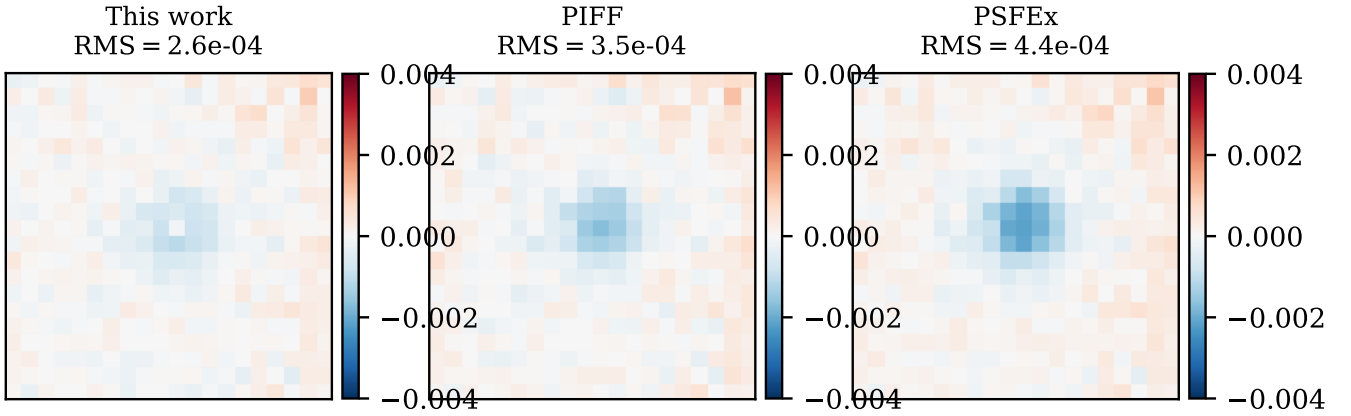
*Blend forward-model*—On a blended simulation, single-star training leaves a  $+4.8\%$  size bias on the star-free

## Reserved-star residuals on the DES test split



**Figure 5.** Reserved-star residual distributions on the DES test split. Our amortized model (red) is the narrowest in size and ellipticity.

## Stacked normalized pixel residual (star – model), 306 reserved stars

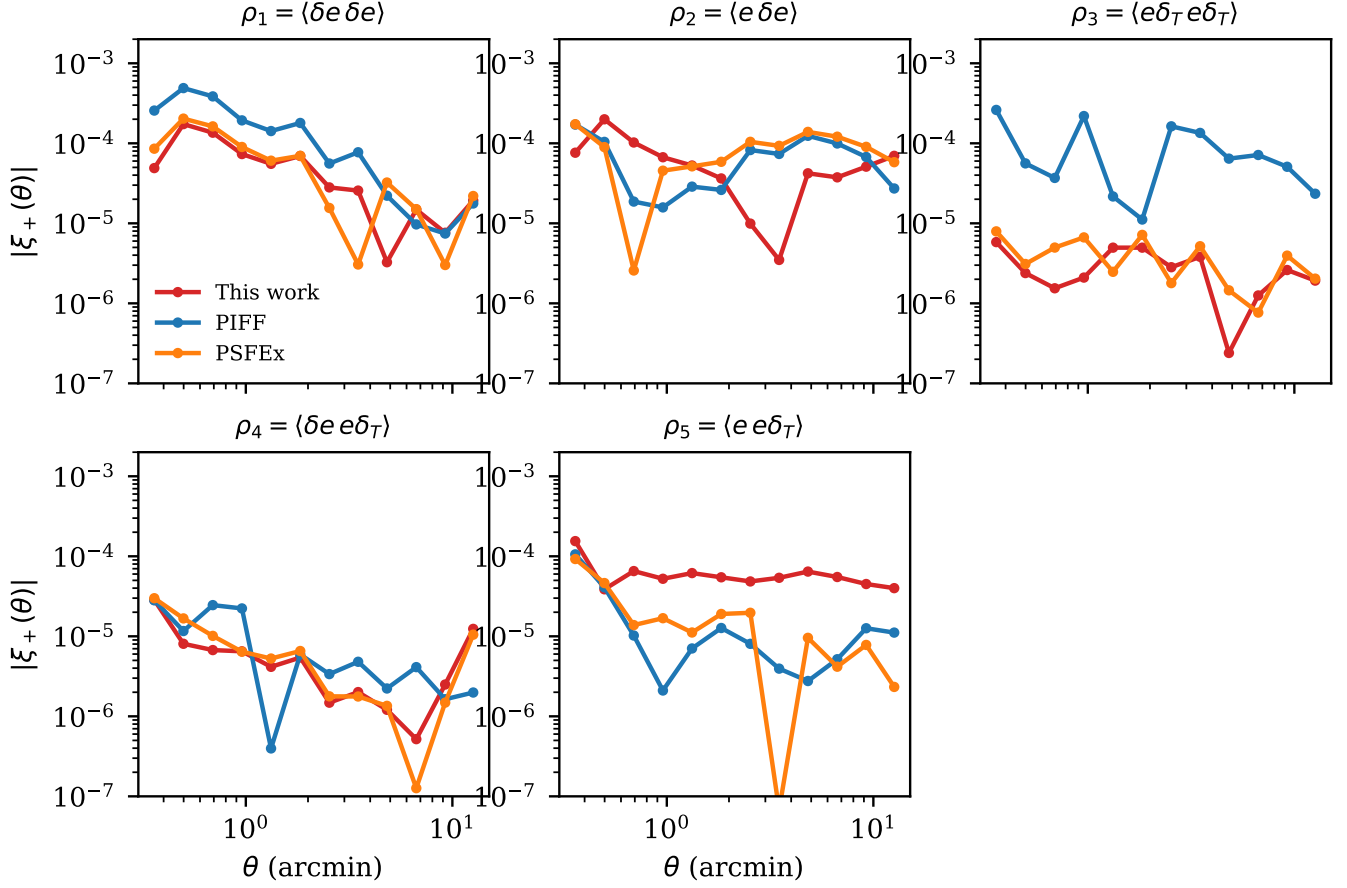


**Figure 6.** Stacked normalized pixel residual (observed star – model, unit-sum) over 306 reserved test stars, on a common scale. Our amortized model leaves the flattest residual; the per-exposure baselines retain a coherent central core—a higher-order shape error that second-moment metrics do not capture (§4.2).

truth grid; the blend forward-model loss reduces it to  $+0.7\%$  ( $\approx 7\times$ ).

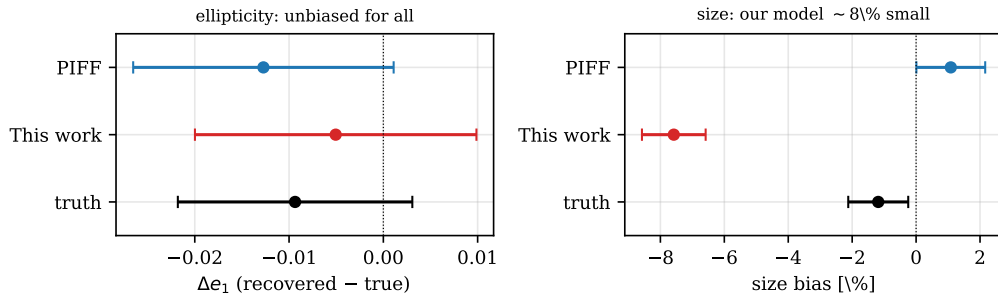
*Galaxy-context corrupts PSF ellipticity—and the fix*—Injecting galaxy detections into the simulation, a model that attends to *all* detections (including galaxies) suffers a clean, reproducible failure: it cannot represent one spin-2 ellipticity component [ $\text{corr}(e_1) = 0.05$  under the polar parametrization; the failure flips to  $e_2$  under a Cartesian parametrization, identifying a rep-

resentation degeneracy rather than noise], while recovering the other perfectly. We localize the cause by ablation: it is *not* data volume, per-source S/N, galaxy size or color, or the  $\chi^2$  cap—the model is treating extended galaxy cutouts as PSF evidence. Restricting the attention context to point sources recovers *both* components completely [ $\text{corr}(e_1, e_2) = 0.98$ , matching the galaxy-free ideal]; a coordinate-system change does not. Galaxies are clearly distinguishable (cutout second moment  $T \approx 33$  versus  $15 \text{ px}^2$  for stars), so this is the model

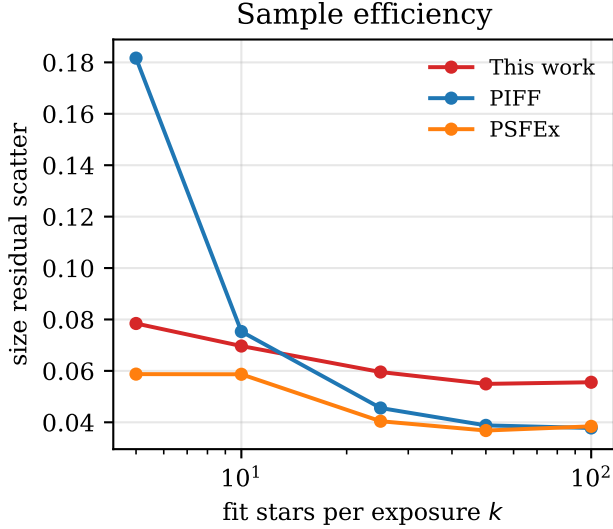
PSF residual  $\rho$ -statistics (reserved stars, all bands)

**Figure 7.** PSF residual  $\rho$ -statistics on reserved stars, per method. Our model (red) is at or below both baselines on  $\rho_1$ – $\rho_4$ ; only  $\rho_5$  (the shape–size coupling) is modestly larger (§4.3).

Galaxy recovery on real data (star-anchored injection)



**Figure 8.** Galaxy recovery on real data (star-anchored injection), median  $\pm$  bootstrap 95% CI per arm. Left: recovered ellipticity is unbiased ( $|\Delta_{e1}| \lesssim 0.004$ ) for all three methods. Right: the methods separate on recovered *size*—our model runs  $\sim 8\%$  small (the PSF-core under-concentration), while PIFF and the truth arm are near zero (§4.4).



**Figure 9.** Size-residual scatter versus the number of fit stars  $k$ . The amortized model is nearly flat in  $k$ ; PIFF degrades sharply below  $k \approx 25$ .

failing to use available information, fixed by the principled restriction that only point sources inform a PSF. On the real test set this restriction carries no accuracy cost: a point-source-context model matches the headline model’s reserved-star residuals (size scatter 0.031 versus 0.030; identical median ellipticity error and shape correlation), confirming that the fix generalizes from simulation to data without trading away real-world performance.

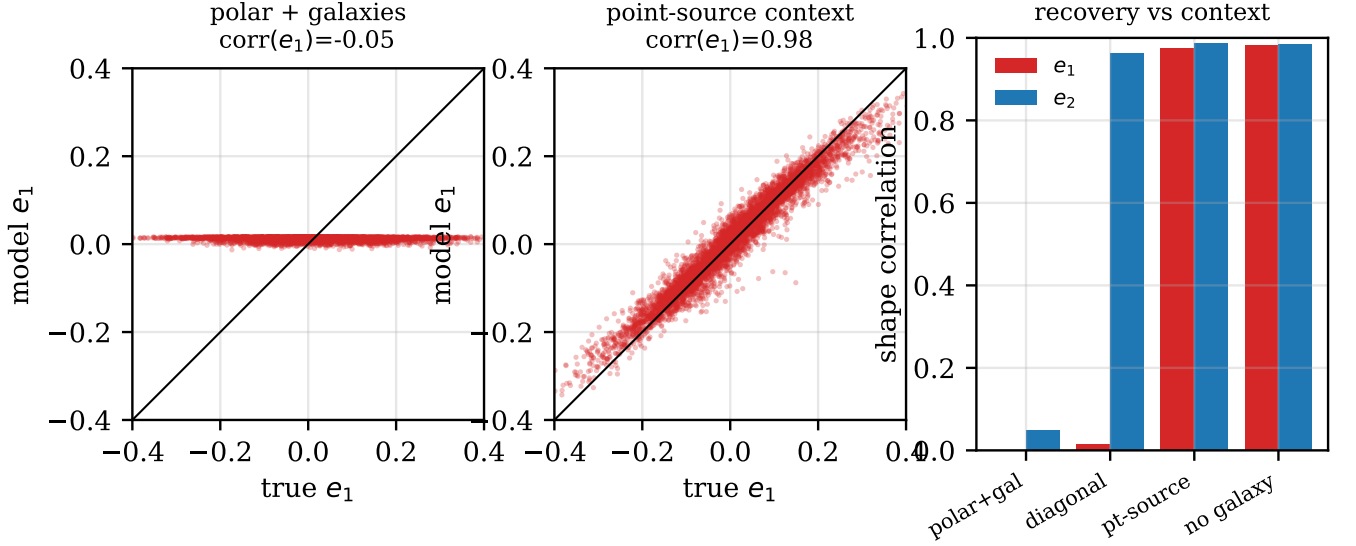
*Seed regularization: EMA versus ensembling*—Amortized estimators carry seed-to-seed prediction noise. On the star-free truth grid we compare a single model, an exponential moving average (EMA) of one model’s weights, and a 4-seed deep ensemble (averaging predicted stamps), each over matched 120-epoch runs (Table 4). EMA gives a small, consistent gain ( $T$ -scatter  $-5\%$ , ellipticity  $\lesssim 2\%$ ); the ensemble is much larger ( $-12\%$  size,  $-24\%$  ellipticity). EMA is therefore *not* a substitute for ensembling—it captures only a fraction of the benefit. The per-seed aggregate scatter is itself small (the model is seed-stable), but the per-position predictions are diverse, which only ensembling exploits. On real reserved stars the ensemble improves the metrics by only  $\sim 1\%$ : that residual is measurement-noise- and blend-limited (corr $\sim 0.66$ , not the truth-grid 0.99), so it masks the model-error reduction that ensembling delivers at the star-free positions where galaxies are measured.

**Table 4.** Truth-grid PSF recovery (4 matched 120-epoch seeds): seed-noise scatter for a single model, EMA, and a 4-seed ensemble.

|                        | $\sigma(\delta e_1)$ | $\sigma(\delta e_2)$ | $\sigma(\delta T/T)$ |
|------------------------|----------------------|----------------------|----------------------|
| single model           | 0.0094               | 0.0099               | 0.0175               |
| EMA (1 $\times$ )      | 0.0092               | 0.0096               | 0.0167               |
| ensemble (4 $\times$ ) | <b>0.0073</b>        | <b>0.0075</b>        | <b>0.0154</b>        |

## 5. DISCUSSION

The amortized field trades a per-exposure fit for a single trained model. Benefits: no test-time fitting, robustness in star-sparse exposures, conditioning on color (chromatic PSF) and flux (brighter-fatter), and a directly forward-modelable output—we predict the pixel-convolved effective PSF, the identifiable quantity the data constrain and the exact kernel a galaxy model is convolved with (§1), so the PSF feeds a shear pipeline with no intervening pixel-deconvolution assumption. Caveats: (i) unlike per-exposure fitters, the amortized model requires many exposures to learn the field (in simulation, ellipticity recovery is unstable below a few thousand exposures; real DES is well above this); (ii) the simulation’s idealized PSF (Moffat, low-order polynomial field) favors the per-exposure baselines, so absolute-accuracy comparisons rest on real data; and (iii) the galaxy-context finding argues that a PSF estimator’s context should be gated to point sources, which we adopt. (iv) The model slightly *under-concentrates* the PSF core: rendered at a fiducial flux its second-moment size is accurate, but it carries  $\sim 2\%$  less encircled energy in the central few pixels than the empirical PSF, which biases recovered galaxy sizes  $\sim 8\%$  low (§4.4). We attribute this to spectral bias of the coordinate decoder—MLP decoders preferentially fit smooth functions and under-resolve sharp peaks. We verify this is the cause and is tractable: widening the decoder’s Fourier frequency basis nearly eliminates the core deficit ( $\sim 2\% \rightarrow 0.1\%$  encircled-energy residual at  $r=2$  px) and cuts the galaxy-size bias by  $\sim 30\%$  (to  $\sim 5\%$ ), at a small cost to reserved-star ellipticity scatter—so a clean fix that improves the core without that cost (a tuned frequency schedule, or sinusoidal-activation decoders) is a tractable refinement rather than an open question. It is the main residual systematic to address before a shear analysis. A separate, milder effect appears only in the reserved-star metric (Fig. 14): evaluated at each star’s *own* flux,  $\delta T/T$  rises to  $\sim +1.2\%$  at faint magnitudes, a spurious flux-size trend in the conditioning; because the



**Figure 10.** Galaxy detections in the attention context corrupt the recovered PSF ellipticity. Left: with galaxies in context, model  $e_1$  collapses to a constant ( $\text{corr} = -0.04$ ). Middle: restricting the context to point sources recovers it ( $\text{corr} = 0.98$ ). Right: shape correlations across configurations—point-source context and galaxy-free both recover *both* components; a coordinate change (diagonal) does not.

inferential PSF is rendered at a fiducial flux, this does not propagate to galaxy measurement.

## 6. CONCLUSION

A single attention-based continuous PSF field, trained across DES single-epoch exposures and applied with no per-exposure fitting, matches or exceeds PIFF and PSFEX on real reserved stars and recovers galaxy ellipticity without coherent bias—on par with PIFF—on a star-anchored injection test, while additionally correcting a chromatic systematic and remaining robust in star-sparse exposures. Controlled simulations motivate two training choices—a blend forward-model loss and point-source-only attention context—the latter from a clean diagnosis of a spin-2 representation failure triggered by galaxy detections in context.

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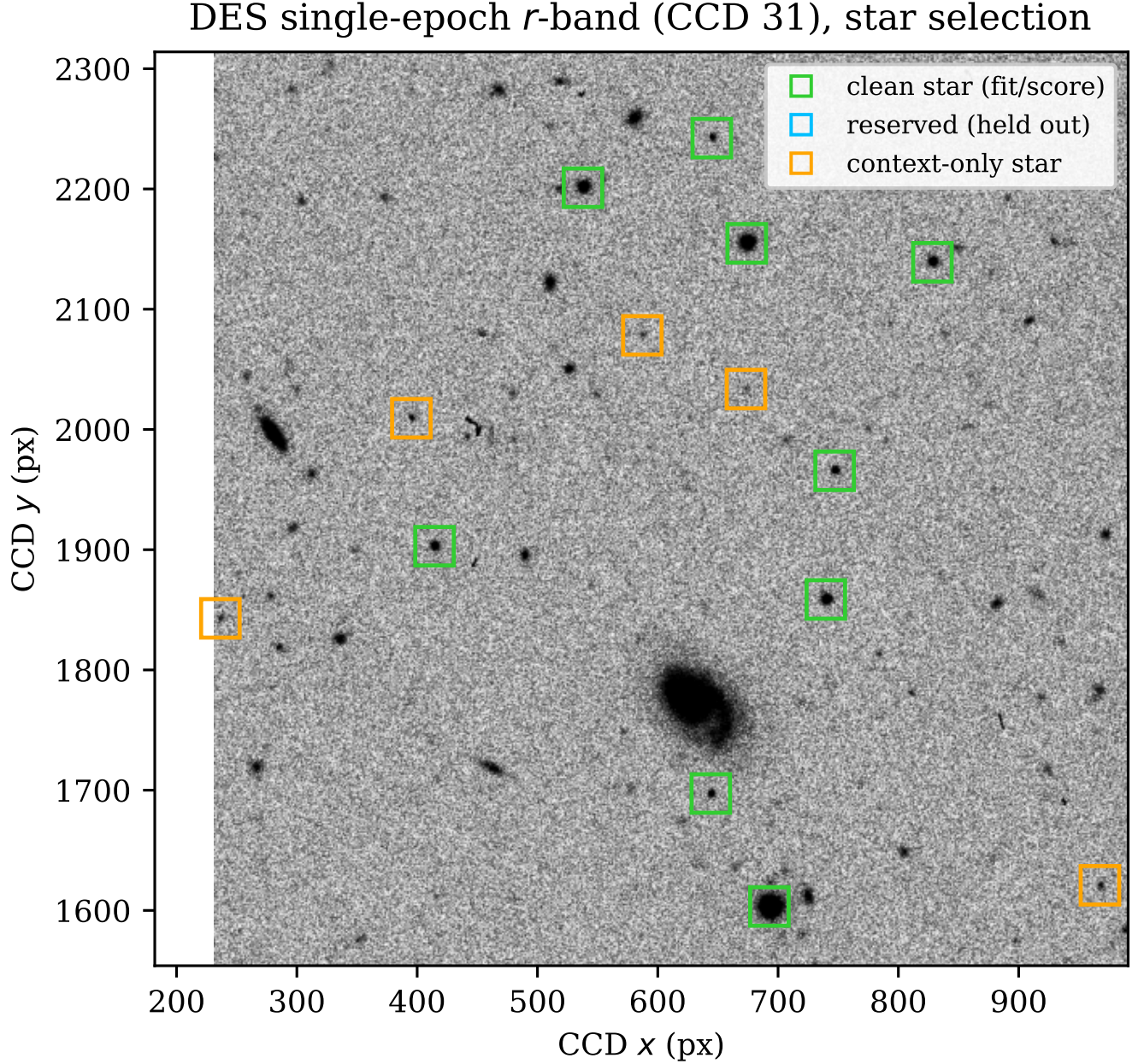
This paper has undergone internal review in the LSST Dark Energy Science Collaboration.

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Baselines: PIFF (PixelGrid, second-order BasisPolynomial,  $4\sigma$  rejection) and PSFEX (PSF\_SAMPLING 0.5,  $45 \times 45$ , PSFVAR\_DEGREES 2; rendered via GALSIM DES\_PSFEX, method='no\_pixel').

*Software:* Astropy (Astropy Collaboration 2022), GalSim (B. T. P. Rowe et al. 2015), SEP (K. Barbary 2016), PyTorch (A. Paszke et al. 2019), PIFF (M. Jarvis et al. 2021), PSFEX (E. Bertin 2011)

## APPENDIX

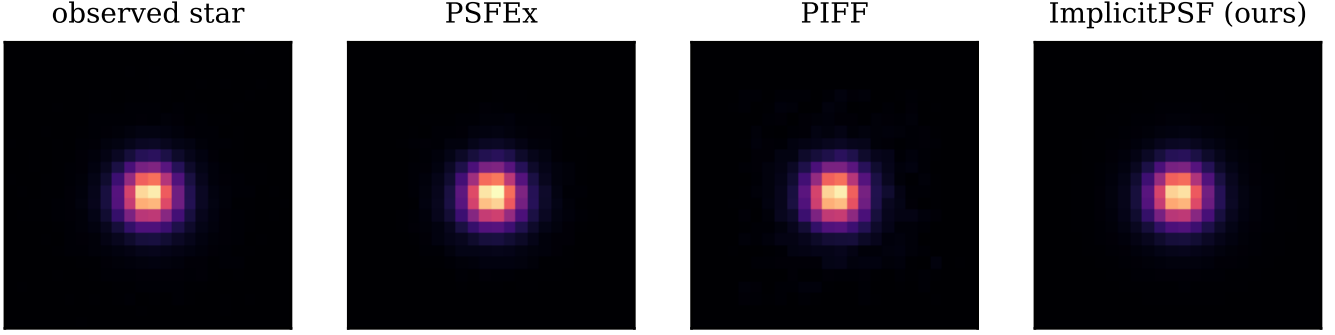


**Figure 11.** Star selection on a real DES exposure. Green: clean stars (fit and score). Blue: reserved (held out for evaluation). Orange: context-only stellar detections. Extended sources (galaxies) are excluded from the attention context.

#### A. REAL-DATA STAR SELECTION AND PSF EXAMPLES

Figure 11 shows a region of a real DES single-epoch *r*-band exposure with the source selection overlaid: clean stars used for fitting and scoring (green), held-out reserved stars (blue), and additional context-only stellar detections (orange). Galaxies (the extended sources, e.g. the bright object near the center) are detected but, under the point-source-context policy (§2), are not attended to. Figure 12 compares, for a single high-S/N reserved test star, the observed stamp against the PSFEx, PIFF, and ImplicitPSF models evaluated at that position (each method having seen only the non-reserved stars). All three reproduce the stellar profile; PIFF’s per-exposure interpolation is visibly noisier in the wings.

Same reserved test star: observed versus modeled PSF



**Figure 12.** The same reserved test star (left, observed) and the PSF predicted at its position by PSFEx, PIFF, and ImplicitPSF. Each model was fit/conditioned on the non-reserved stars only. Common linear stretch.

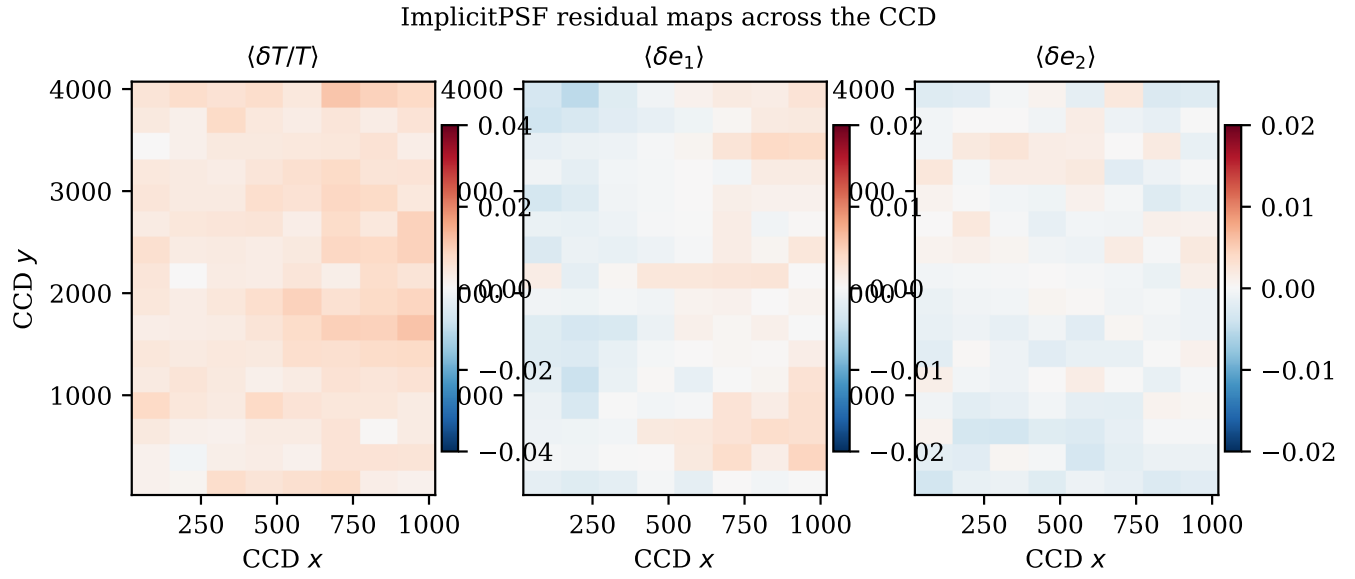
**Table 5.** Reserved-star residuals by band (test split).  $T$  scatter is the robust standard deviation of  $\delta T/T$ ;  $|\delta e_1|$  is the median absolute  $e_1$  residual; corr is the model–star  $e_1$  correlation. Best per band in bold.

| Band | Method           | $T$ scat      | $ \delta e_1 $ | corr( $e_1$ ) |
|------|------------------|---------------|----------------|---------------|
| $g$  | <b>This work</b> | <b>0.0331</b> | <b>0.0164</b>  | <b>0.475</b>  |
|      | PIFF             | 0.0439        | 0.0188         | 0.448         |
|      | PSFEx            | 0.0417        | 0.0173         | 0.473         |
| $r$  | <b>This work</b> | <b>0.0291</b> | <b>0.0144</b>  | <b>0.576</b>  |
|      | PIFF             | 0.0366        | 0.0164         | 0.535         |
|      | PSFEx            | 0.0340        | 0.0151         | 0.572         |
| $i$  | <b>This work</b> | <b>0.0274</b> | <b>0.0136</b>  | <b>0.643</b>  |
|      | PIFF             | 0.0342        | 0.0155         | 0.605         |
|      | PSFEx            | 0.0339        | 0.0144         | 0.637         |
| $z$  | <b>This work</b> | <b>0.0272</b> | <b>0.0140</b>  | <b>0.601</b>  |
|      | PIFF             | 0.0387        | 0.0160         | 0.503         |
|      | PSFEx            | 0.0359        | 0.0152         | 0.589         |
| $Y$  | <b>This work</b> | <b>0.0391</b> | <b>0.0210</b>  | <b>0.568</b>  |
|      | PIFF             | 0.0513        | 0.0226         | 0.523         |
|      | PSFEx            | 0.0449        | 0.0221         | 0.550         |

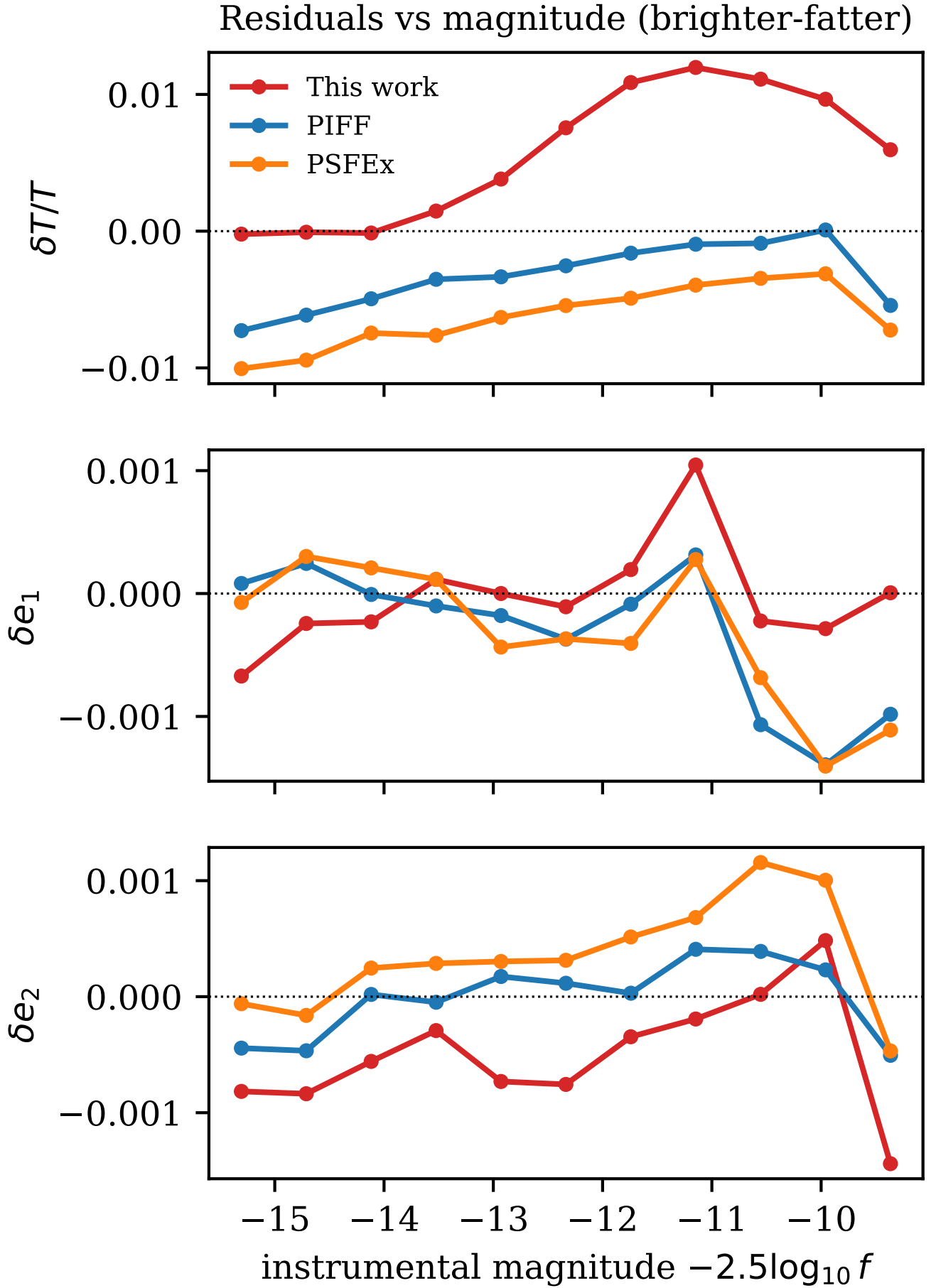
## B. BENCHMARK VALIDATION DIAGNOSTICS

For comparison with standard per-exposure PSF validation (cf. [M. Jarvis et al. 2021](#)), we report the same diagnostics.

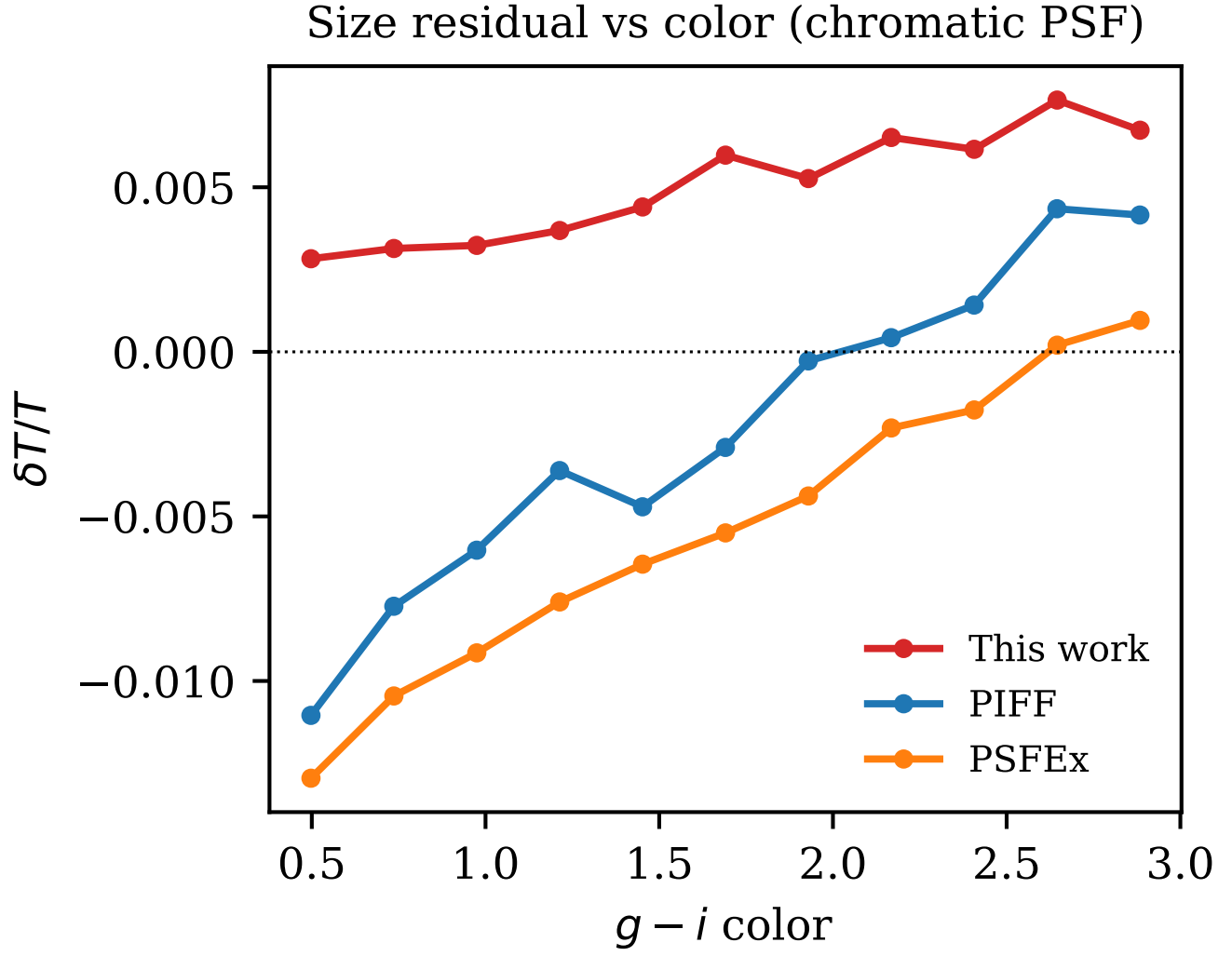
Table 5 breaks the headline metrics down by band. The amortized model has the smallest size scatter and median ellipticity error in every band, with the largest margins in  $Y$  and  $z$ , where the broader, more variable PSF is hardest for per-exposure fits.



**Figure 13.** Median ImplicitPSF residuals binned across the CCD (cf. PIFF Fig. 9).

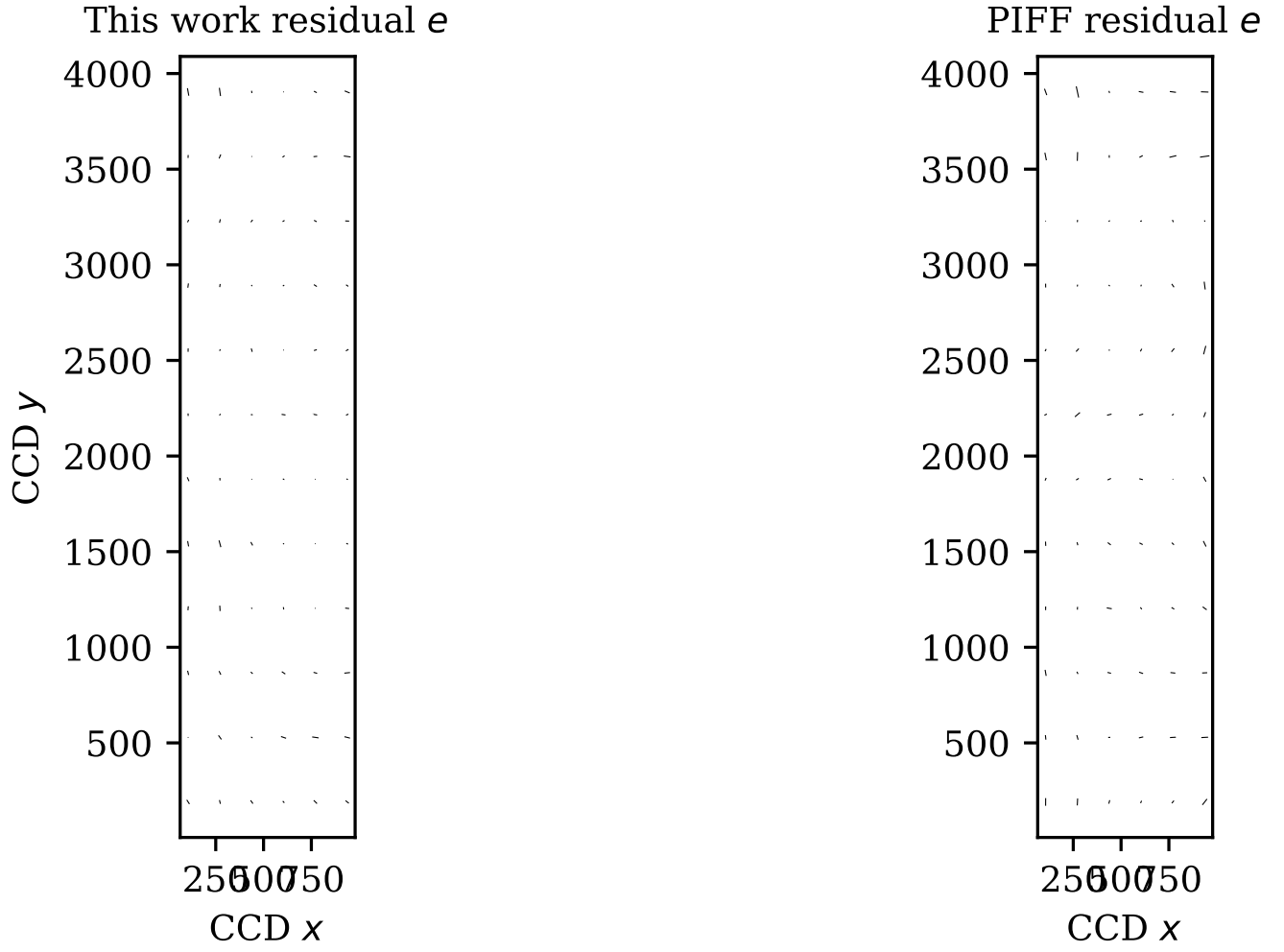


**Figure 14.** Residuals versus instrumental magnitude (cf. PIFF Fig. 10). Our model (red) carries a magnitude-dependent size residual peaking at  $+1.2\%$  (85).



**Figure 15.** Size residual versus  $g - i$  color (cf. PIFF Fig. 11); color conditioning keeps our chromatic residual flat.

## Residual PSF ellipticity whiskers



**Figure 16.** Residual PSF-ellipticity whiskers across the CCD, ImplicitPSF and PIFF (cf. PIFF Fig. 2).

## REFERENCES

- Anderson, J., & King, I. R. 2000, *PASP*, 112, 1360, doi: [10.1086/316632](https://doi.org/10.1086/316632)
- Astropy Collaboration. 2022, *ApJ*, 935, 167, doi: [10.3847/1538-4357/ac7c74](https://doi.org/10.3847/1538-4357/ac7c74)
- Barbary, K. 2016, *JOSS*, 1, 58, doi: [10.21105/joss.00058](https://doi.org/10.21105/joss.00058)
- Bernstein, G. M., & Jarvis, M. 2002, *AJ*, 123, 583, doi: [10.1086/338085](https://doi.org/10.1086/338085)
- Bertin, E. 2011, *ADASS XX*, 442, 435
- Jarvis, M., et al. 2016, *MNRAS*, 460, 2245
- Jarvis, M., et al. 2021, *MNRAS*, 501, 1282, doi: [10.1093/mnras/staa3679](https://doi.org/10.1093/mnras/staa3679)
- Lauer, T. R. 1999, *PASP*, 111, 227, doi: [10.1086/316319](https://doi.org/10.1086/316319)
- Liaudat, T. I., Starck, J.-L., & Kilbinger, M. 2023, *Frontiers in Astronomy and Space Sciences*, 10, 1158213, doi: [10.3389/fspas.2023.1158213](https://doi.org/10.3389/fspas.2023.1158213)
- Paszke, A., et al. 2019, in *NeurIPS*
- Rowe, B. 2010, *MNRAS*, 404, 350
- Rowe, B. T. P., et al. 2015, *Astronomy and Computing*, 10, 121, doi: [10.1016/j.ascom.2015.02.002](https://doi.org/10.1016/j.ascom.2015.02.002)
- Vaswani, A., Shazeer, N., Parmar, N., et al. 2017, in *Advances in Neural Information Processing Systems* (NeurIPS), Vol. 30